1 THE PROBLEM → THE SYSTEM

From inconsistent lubricant application to intelligent, data-driven process control

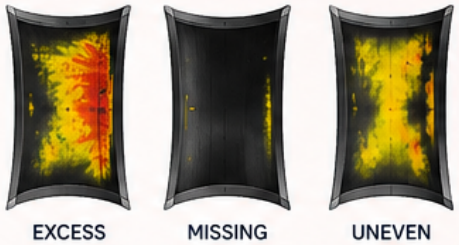
THE PROBLEM

Lubricant distribution inside the tyre is hard to see, varies across tyres, and impacts quality.



1. LUBRICANT APPLICATION

Lubricant is sprayed on the inner surface during manufacturing.



2. INCONSISTENT DISTRIBUTION

Distribution can be uneven – too much, too little, or in the wrong areas.

- QUALITY ISSUES
- HIGH REWORK & WASTE
- CUSTOMER DISSATISFACTION

3. IMPACT

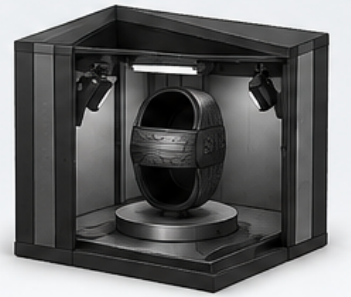
Leads to quality issues, rework, customer complaints, and waste.



4. THE REAL QUESTION

What is the true lubricant distribution? Why is it happening? Is the process drifting over time?

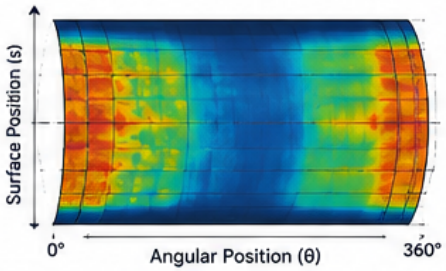
OUR SYSTEM: 3 LAYERS OF INTELLIGENCE BUILT ON A COMMON FOUNDATION



COMMON FOUNDATION

Physical Inspection System

Controlled cameras, lighting, rotation and encoder capture the complete inner surface.



COMMON FOUNDATION

Digital Surface Model (θ, s)

The inner surface is reconstructed in a common coordinate system for every tyre.

IDEA 1

Spray Fingerprint (Normality)



What should a good tyre's lubricant distribution look like, and how much does this tyre deviate from it?

IDEA 2

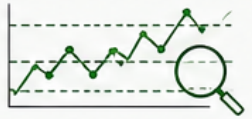
Process-Aware Expectation



Given what the machine was doing, what lubricant distribution should we expect, and does the observed distribution agree with it?

IDEA 3

Drift & Root-Cause Intelligence



Is the manufacturing process gradually deteriorating, and what could be causing it?

OUR GOAL Turn what is invisible into measurable intelligence – detect issues, understand causes, and prevent defects through data-driven insights.

2 THE TYRE INSPECTION STATION

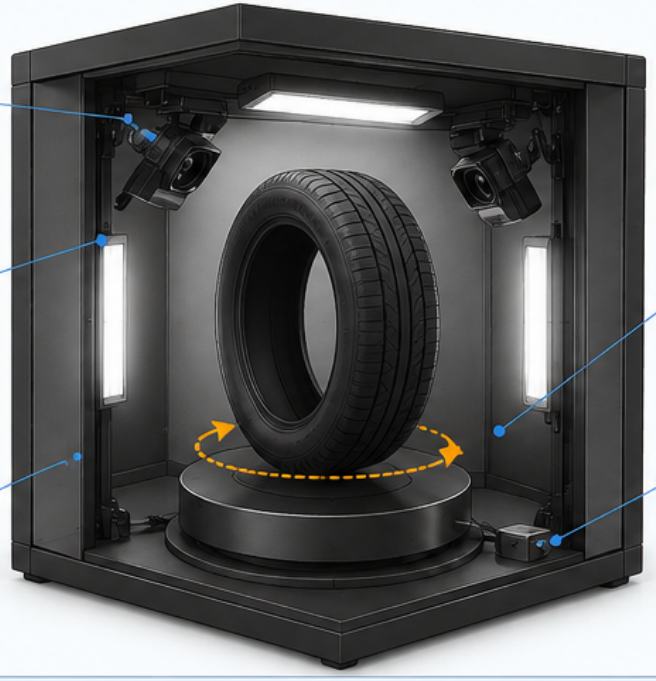
Controlled. Synchronized. Complete.

[COMMON] STEP 1 PHYSICAL INSPECTION & DATA ACQUISITION

INDUSTRIAL CAMERAS
High-resolution cameras capture the inner surface from fixed positions.

CONTROLLED ILLUMINATION
Uniform, stable lighting ensures consistent images and minimizes reflections.

ENCLOSED CHAMBER
Blocks external light and environmental variations for reliable capture.



CONTROLLED ROTATION
The tyre rotates slowly and smoothly for full 360° coverage of the inner surface.

ROTATION ENCODER
High-precision encoder provides the exact angular position θ(t) at every instant.

KEY PRINCIPLE: The tyre rotates. Cameras stay fixed.
The encoder tells us exactly which angular position is being observed at every instant.

360° COMPLETE INNER SURFACE CAPTURE



WHAT WE GET

SYNCHRONIZED IMAGES
Images are time-stamped and tagged with angular position θ.

ANGULAR POSITION (θ)
Encoder provides precise angle for every captured frame/line.

COMPLETE DATASET
Full inner surface data with exact spatial reference.

FLEXIBLE CAMERA IMPLEMENTATION

AREA-SCAN SETUP
(Multiple Cameras)



Faster coverage with multiple area-scan cameras.

OR

LINE-SCAN SETUP
(Single Camera + Motion)



High-resolution line-scan with synchronized rotation.

BOTH APPROACHES LEAD TO THE SAME OUTPUT: (θ, s) SURFACE DATA

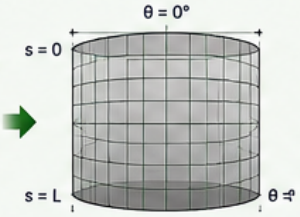
FROM ROTATION TO COORDINATE



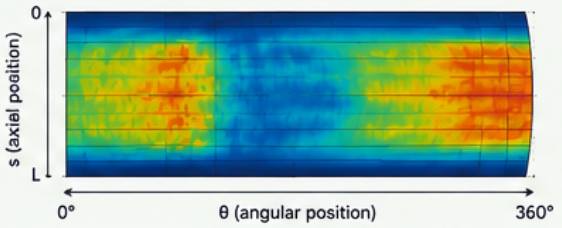
TYRE ROTATES
Continuous rotation during inspection.



ENCODER READS ANGLE θ(t)
Every image/line is mapped to an angle.



MAP TO (θ, s)
Unfold the inner surface into a cylindrical coordinate system.



COMPLETE INNER SURFACE MAP
A full 2D map of the inner surface in (θ, s) coordinates.

THE OUTCOME: A controlled, synchronized system that captures the entire inner surface with precise spatial reference — the foundation for all intelligent analysis.

✓ Controlled ✓ Repeatable ✓ Traceable

3 FROM CAMERA IMAGES → DIGITAL TYRE SURFACE

Transform raw observations into a precise, comparable representation (θ, s)

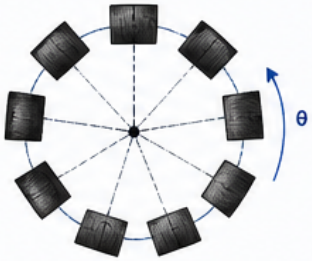
1 RAW CAMERA OBSERVATIONS

Cameras capture the inner surface as the tyre rotates.



2 GEOMETRIC REGISTRATION

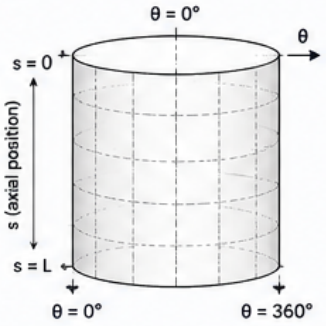
Each image is mapped to the correct angular position using the encoder and system calibration.



Every pixel is assigned a precise angle θ.

3 COORDINATE SYSTEM (θ, s)

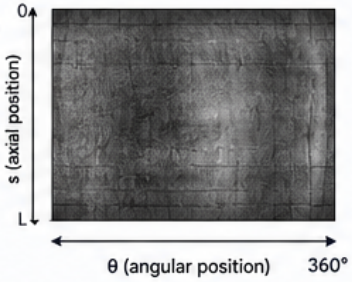
The inner surface is unrolled into a cylindrical coordinate system.



θ : Angular position (0°–360°)
s : Axial position (0 to L)

4 UNROLL & RESAMPLE (MAP)

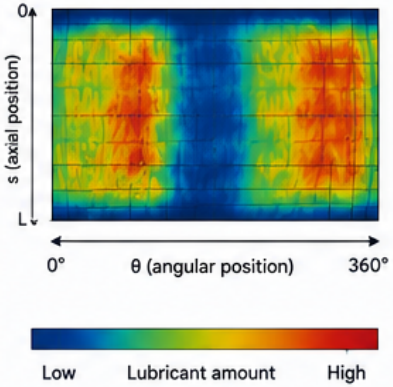
The cylindrical surface is unrolled and resampled to a uniform grid to create the digital surface.



Result: Canonical surface map in (θ, s) coordinates.

5 CANONICAL SURFACE MAP (θ, s)

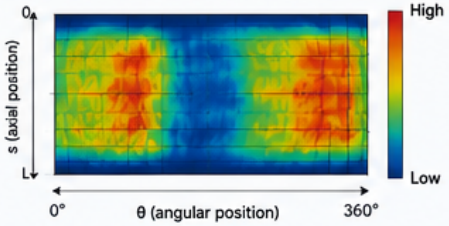
A consistent, comparable digital representation of the entire inner surface.



WHAT WE GENERATE

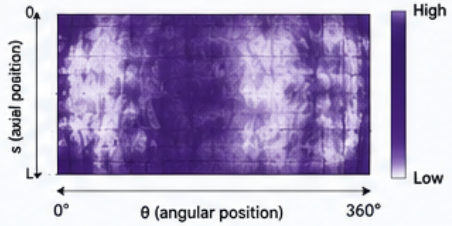
1 OBSERVATION MAP

Measured lubricant distribution on the inner surface.



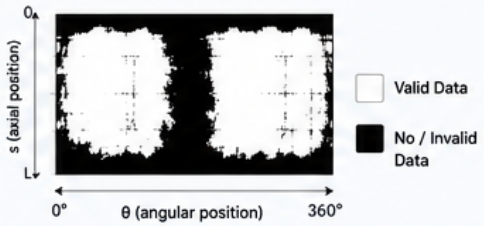
2 CONFIDENCE MAP

Reliability of each measurement (affected by lighting, occlusion, etc.).



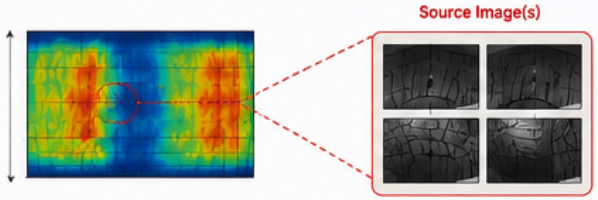
3 VALIDITY / MASK

Regions with valid data vs. occluded or unreliable areas.



TRACEABILITY TO RAW EVIDENCE

Every cell in the surface map is linked back to the original image(s) that produced it.



This ensures auditability, explainability, and engineering trust.

KEY TAKEAWAY

We do not compare raw images.

We compare normalized, coordinate-registered surfaces in (θ, s).

WHY IT MATTERS

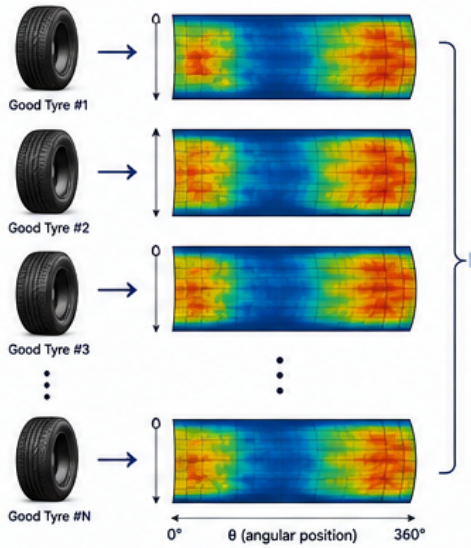
- Enables accurate spatial comparison
- Removes camera and lighting bias
- Makes data consistent across time & lines
- Foundation for all three intelligence layers

4 THE TYRE LEARNS WHAT "NORMAL" LOOKS LIKE

IDEA 1: Normal-Process / Spray-Fingerprint Modelling

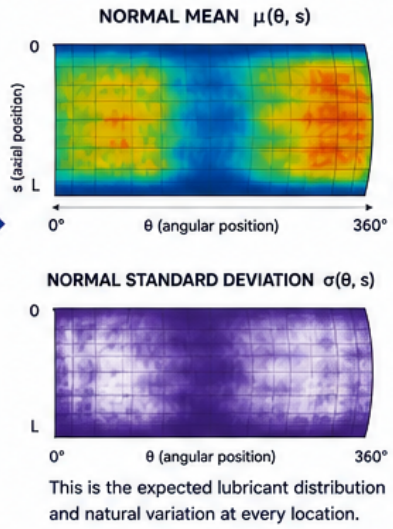
1 LEARN FROM GOOD TYRES

Collect many good (accepted) tyres and build their surface maps.



2 BUILD THE NORMAL SPRAY FINGERPRINT

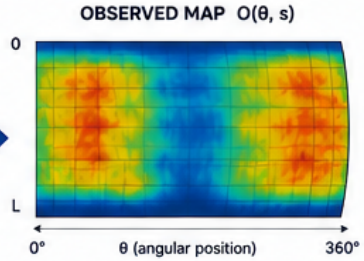
Aggregate all good tyres to capture the true normal pattern.



This is the expected lubricant distribution and natural variation at every location.

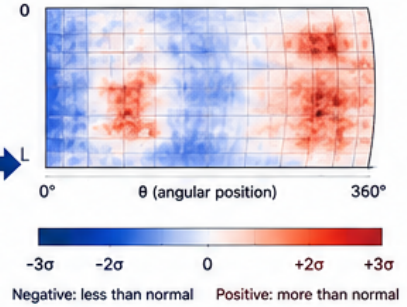
3 INSPECT A NEW TYRE

Capture and reconstruct the new tyre's surface map.



4 COMPARE TO NORMAL

Quantify how much the new tyre deviates from normal.



OVERALL TYRE HEALTH SCORE

Example: 78 / 100

5 DECISION

Classify the tyre based on deviation and severity.

ACCEPT (NORMAL)

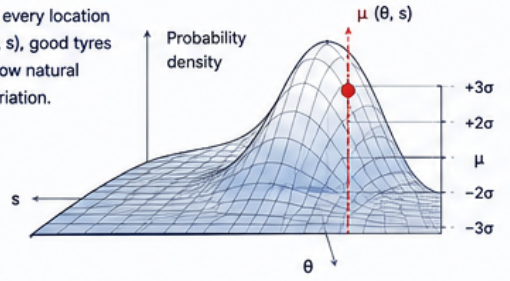
Deviations are within natural variation.

REJECT (ABNORMAL)

Significant abnormal distribution detected.

WHAT DOES "NORMAL" MEAN?

At every location (θ, s), good tyres show natural variation.

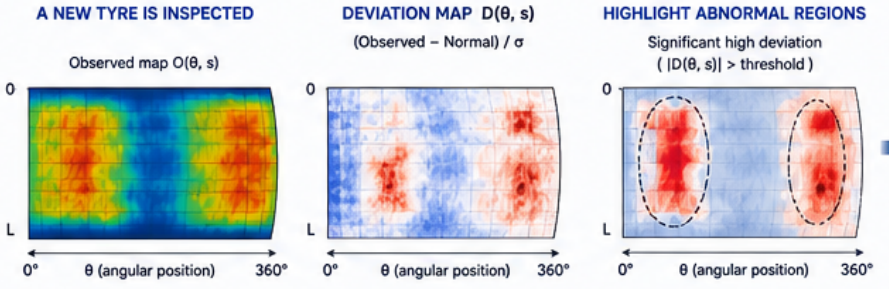


NORMAL MODEL AT EACH LOCATION

- μ(θ, s) → Expected (mean) lubricant amount
- σ(θ, s) → Natural variation (standard deviation)

The model knows what is typical and how much variation is OK.

FROM LOCAL DEVIATION TO ACTIONABLE INSIGHT



INSIGHTS

- Where is excess lubricant?
- Where is insufficient lubricant?
- How large and severe is it?
- How does it affect quality?

KEY TAKEAWAYS

- Built from many good tyres
- Captures normal pattern (μ)
- Understands natural variation (σ)
- Finds real abnormalities
- Objective, spatial and explainable

WHY THIS MATTERS

- ☑ No fixed global threshold.
- ☑ Respects spatial variation.
- ☑ Consistent decisions across tyres and time.
- ☑ Foundation for process-aware and drift analysis.

5 FROM "DEFECT" TO "WHY?"

IDEA 2: Physics + Process Parameters + Vision

Question it answers: Given what the machine was doing, what lubricant distribution should we expect, and **does the observed distribution agree with it?**

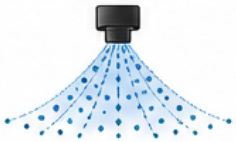
1 PROCESS STATE
What the machine was doing

- **Nozzle Pressure** P = 6.5 bar
- **Flow Rate** Q = 120 ml/min
- **Nozzle Position** Offset = +3 mm
- **Tyre Rotation** ω = 40 rpm
- **Spray Duration** t = 2.5 s

Live process parameters captured for this tyre.

2 PHYSICS + GEOMETRY
Understand the system

SPRAY PHYSICS



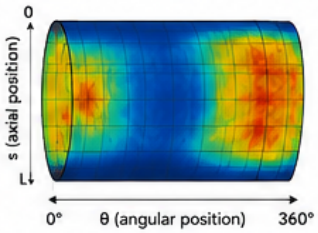
TYRE GEOMETRY



Physics of spray + inner surface shape determine how lubricant deposits across (θ, s).

3 EXPECTED DEPOSITION
What should happen

PROCESS MODEL
(Physics-based / Data-driven)



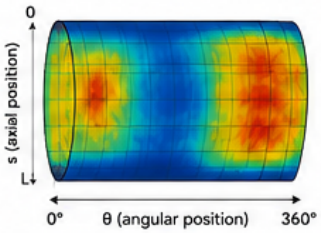
EXPECTED MAP E(θ, s)



Given the process state and tyre geometry, this is the lubricant distribution we expect.

4 OBSERVED DEPOSITION
What actually happened

FROM INSPECTION SYSTEM
(Digital Surface Model)



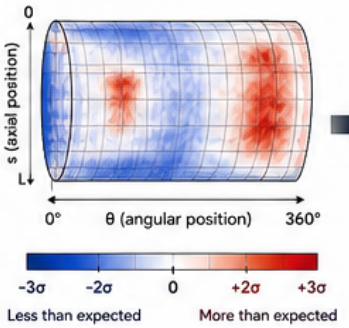
OBSERVED MAP O(θ, s)



Actual lubricant distribution measured on the tyre.

5 COMPARE
Do they agree?

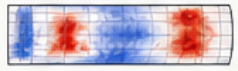
DEVIATION MAP D(θ, s) = O - E



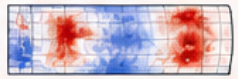
Positive (red): more lubricant than expected. Negative (blue): less lubricant than expected.

6 PROCESS-RELATED DEVIATION

CONSISTENT (EXPLAINED)
Deviation pattern matches what this process state would cause.
Rooted in process behaviour.



INCONSISTENT (UNEXPLAINED)
Deviation cannot be explained by the process state.
Investigate other causes (equipment, environment, wear, etc.).



IDEA 1 vs IDEA 2 — DIFFERENT QUESTIONS, DIFFERENT MODELS

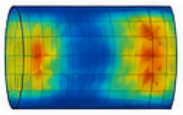
IDEA 1: Normality Model (Historical)
How does this tyre compare to good tyres in the past?

NORMAL FINGERPRINT
μ(θ, s) ± σ(θ, s)



VS

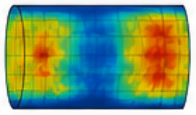
OBSERVED MAP O(θ, s)



Tells us if the tyre looks normal compared to history.
Population-based.

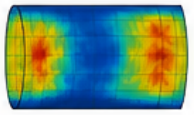
IDEA 2: Process Model (Conditional)
Does the tyre match what the process should produce now?

EXPECTED MAP E(θ, s)



VS

OBSERVED MAP O(θ, s)

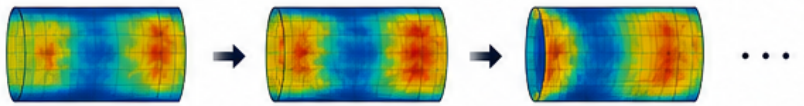


Tells us if the tyre matches what the current process state predicts.
Process-conditioned.

HOW PROCESS PARAMETERS CHANGE THE EXPECTED MAP

Pressure · Flow · Nozzle Offset · Rotation · Duration

Change any parameter → Expected deposition changes



The model lets us predict outcomes before defects happen.

KEY TAKEAWAY

We don't just see the defect.
We understand if the process explains it.
Turning inspection into process understanding.

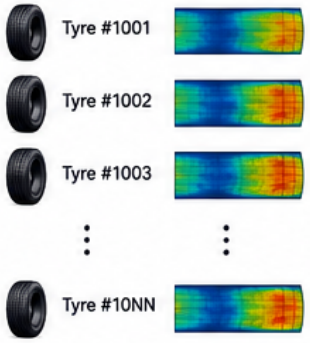
- Links machine settings to actual lubricant patterns
- Separates explained deviation from true abnormalities
- Enables targeted corrective actions
- Builds the bridge to process drift and root cause

6 FROM INDIVIDUAL TYRES → PROCESS INTELLIGENCE

IDEA 3: Process Drift + Root-Cause Intelligence

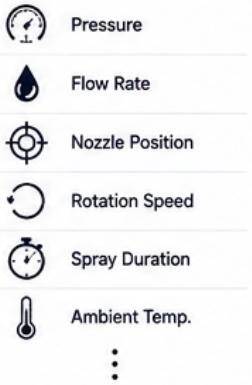
Question it answers:
Is the manufacturing process gradually deteriorating, and what could be causing it?

1 INSPECTION HISTORY
Every tyre is inspected



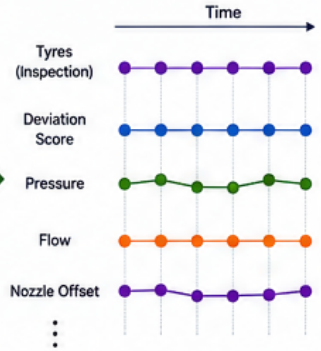
Each tyre → surface map in (θ, s) + deviation metrics

2 PROCESS HISTORY
Machine data over time



Time-stamped parameters for every tyre

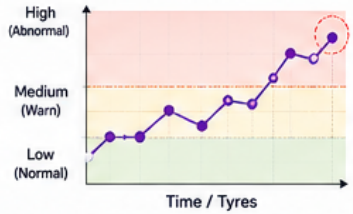
3 COMBINE & ALIGN
Bring data on a timeline



Aligned, synchronized dataset for analysis

4 TRACK OVER TIME
How the process behaves

EXAMPLE: DEVIATION SCORE (D)



EXAMPLE: PRESSURE



Trends reveal slow changes before defects spike

5 EARLY WARNING
Detect drift early



WARNING
Process drift detected

WHY EARLY?
- Catch issues before they create defects
- Prevent waste & rework
- Protect quality & customer satisfaction
- Lower production cost

6 ROOT-CAUSE INTELLIGENCE
Find what is causing it

TOP LIKELY CAUSES

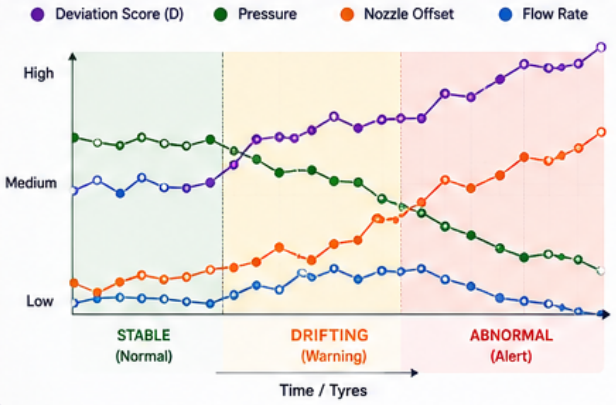
1	Nozzle Position Drift	68%
2	Pressure Drop	21%
3	Flow Instability	7%
4	Nozzle Wear	4%

Data-driven root-cause ranking with confidence score

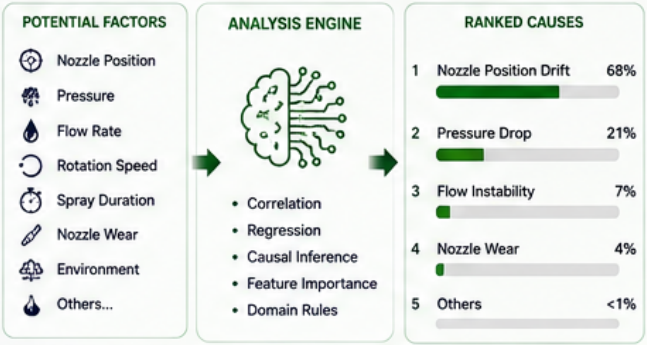
FROM RAW DATA TO INSIGHTS



EXAMPLE: DRIFT OVER TIME



ROOT-CAUSE ANALYSIS PIPELINE



Actionable insights for engineers and process teams

THE OUTCOME

- Prevent defects before they happen
- Reduce waste, rework & cost
- Improve quality consistency
- Increase equipment reliability & uptime
- Data-driven decisions, not guesswork

= **Better Process. Better Tyres. Higher Profitability.**

7 THE COMPLETE INTELLIGENCE LOOP

From Physical Inspection to Actionable Process Intelligence

GOAL: Deliver the right intelligence at the right time to produce better tyres, every day.

IDEA 1: Normality (What is normal?) **IDEA 2:** Process Understanding (Why did it happen?) **IDEA 3:** Process Intelligence (Is it getting worse?)

1 PHYSICAL INSPECTION [COMMON]

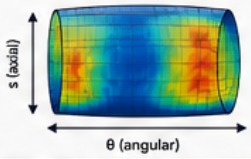
Controlled station captures the complete inner surface while the tyre rotates.



- Cameras
- Controlled Lighting
- Rotation Encoder

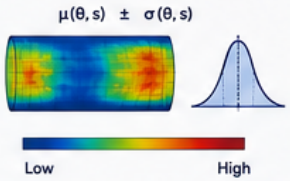
2 DIGITAL SURFACE MODEL (θ, s) [COMMON]

Raw images are transformed into a canonical, traceable surface representation.



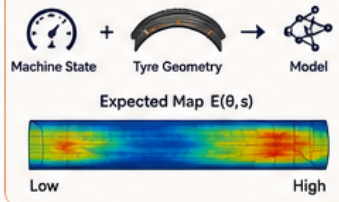
3 NORMAL SPRAY FINGERPRINT [IDEA 1]

Built from many good tyres. Captures the natural variation at every location.



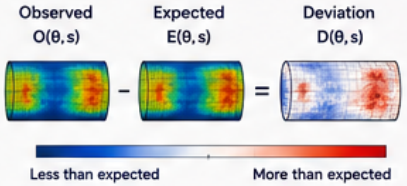
4 PROCESS-AWARE EXPECTATION [IDEA 2]

Given the machine state and tyre geometry, the model predicts the expected deposition.



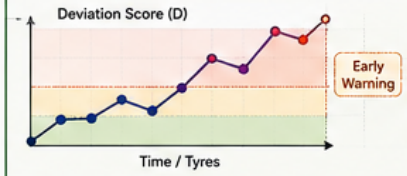
5 DEVIATION ANALYSIS & INTERPRETATION [IDEA 1 + IDEA 2]

Compare observed vs expected and normal to generate meaningful insights.



6 DRIFT & ROOT-CAUSE INTELLIGENCE [IDEA 3]

Historical inspections + process data reveal drift, raise early warnings, and identify probable root causes.



Root Cause Ranking

1	Nozzle Position Drift	68%
2	Pressure Drop	21%
3	Flow Instability	7%
4	Nozzle Wear	4%

ONE SYSTEM. THREE INTELLIGENCES. BETTER DECISIONS.



INTELLIGENCE INTO ACTION

TARGETED ACTIONS
Adjust the right parameter at the right time.

PROCESS IMPROVEMENT
Stabilize the process and reduce variation.

BETTER QUALITY
Fewer defects, more consistent tyres.

HIGHER PERFORMANCE
Lower cost, higher uptime, more profitability.

CLOSED-LOOP LEARNING

Every action changes the process. Every next tyre teaches the system. The model keeps getting smarter.



THE BIG PICTURE

Physical World → Sensing → Digital Surface (θ, s) → **IDEA 1** Normality (What is normal?) + **IDEA 2** Process Understanding (Why did it happen?) + **IDEA 3** Process Intelligence (Is it getting worse?) → Action → Better Tyres

MEASURE PRECISELY. UNDERSTAND DEEPLY. ACT EARLY. DELIVER CONSISTENTLY.

☆ **INTELLIGENCE THAT DRIVES EXCELLENCE.**